

4/2/2025

INTRODUCTION TO SOFTWARE ENGINEERING

ASSIGNMENT # 3



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INTRODUCTION TO SOFTWARE ENGINEERING

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CLASS DIAGRAM, ACITIVITY DIAGRAM ,SEQUENCE DIAGRAM

CLASS DIAGRAM

DESCRIPTION:

Description of the Class Diagram

Classes and Attributes

1. User

- **Attributes:**

- -userID: string
- -userName: string
- -password: string
- -email: string

- **Methods:**

- +login()
- +register()
- +updateProfile()

2. Student (inherits from User)

- **Attributes:**

- -studentID: string
- -viewGrades()

- Methods:
 - +submitAssignment()

3. Instructor (inherits from User)

- Attributes:
 - -instructorID: string
- Methods:
 - +createAssignment()
 - +uploadGrades()
 - +markAttendance()

4. Admin (inherits from User)

- Attributes:
 - -adminID: string
- Methods:
 - +manageSystemSettings()
 - +generateReports()

5. Authentication service

- Attributes:
 - -serviceID: string
- Methods:
 - +authorizeUser()

6. Course

- Attributes:
 - -courseID: string
 - -title: string
 - -description: string
- Methods:
 - +enroll()
 - +drop()

7. Assignment

- Attributes:
 - -assignmentID: string
 - -title: string
 - -deadline: date
- Methods:
 - +upload()

8. Submission

- Attributes:
 - -submissionDate: date
 - -fileType: file
- Methods:
 - +submit()

9. Attendance

- Attributes:
 - -attendanceDate: date
 - -present: boolean

10. LectureNotes

- Attributes:
 - -lectureID: string
 - -fileURL: string
- Methods:
 - +download()

11. Grade

- Attributes:
 - -marks: float
 - -feedback: string

Relationships

1. Inheritance:

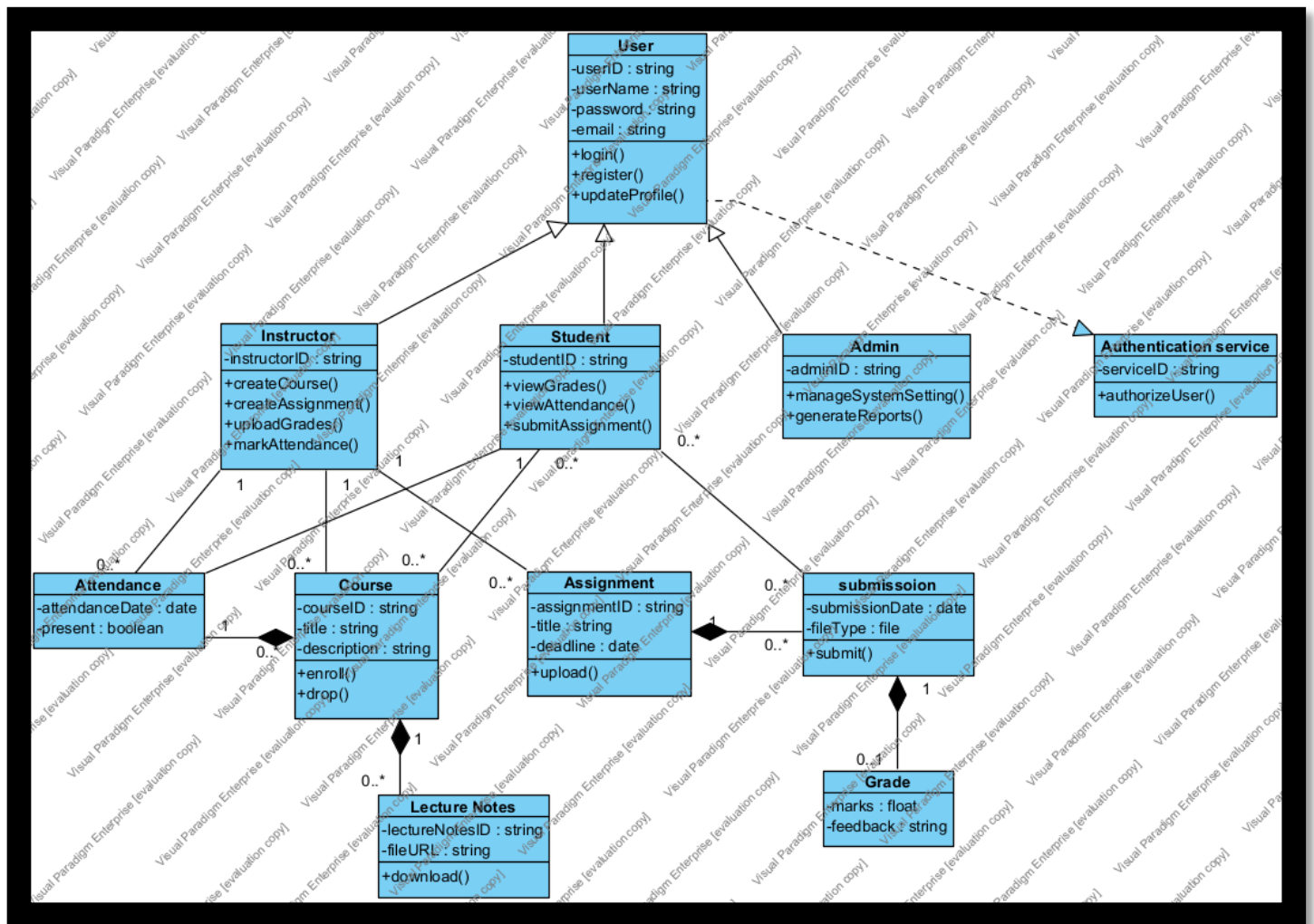
- ❖ User is the parent class, with Student, Instructor, and Admin as subclasses (generalization, indicated by a triangle pointing to User).
- ❖ Authentication service has a dashed line to User (likely an association or dependency for authorization).

2. Associations:

- ❖ Student to Assignment (1 to 0..*): A student can submit multiple assignments.
- ❖ Instructor to Assignment (1 to 0..*): An instructor can create multiple assignments.
- ❖ Student to Course (0..* to 0..*): A student can enroll in multiple courses, and a course can have multiple students.
- ❖ Instructor to Course (1 to 0..*): An instructor can manage multiple courses.
- ❖ Assignment to Submission (1 to 0..*): An assignment can have multiple submissions.
- ❖ Course to LectureNotes (0..* to 0..*): A course can have multiple lecture notes.
- ❖ Assignment to Grade (0..* to 0..*): An assignment can have multiple grades.
- ❖ Instructor to Attendance (1 to 0..*): An instructor can mark attendance for multiple instances.

3. Dependencies:

- ❖ Authentication service to User (dashed arrow): Suggests the service authorizes users.



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ACTIVITY DIAGRAM

1 ACTIVITY DIAGRAM OF LOGIN PAGE

The activity diagram above represents the **Login process** of a user in a system. It shows the step-by-step flow from the moment the user opens the login form to either a successful login or an error message.

Explanation of Steps:

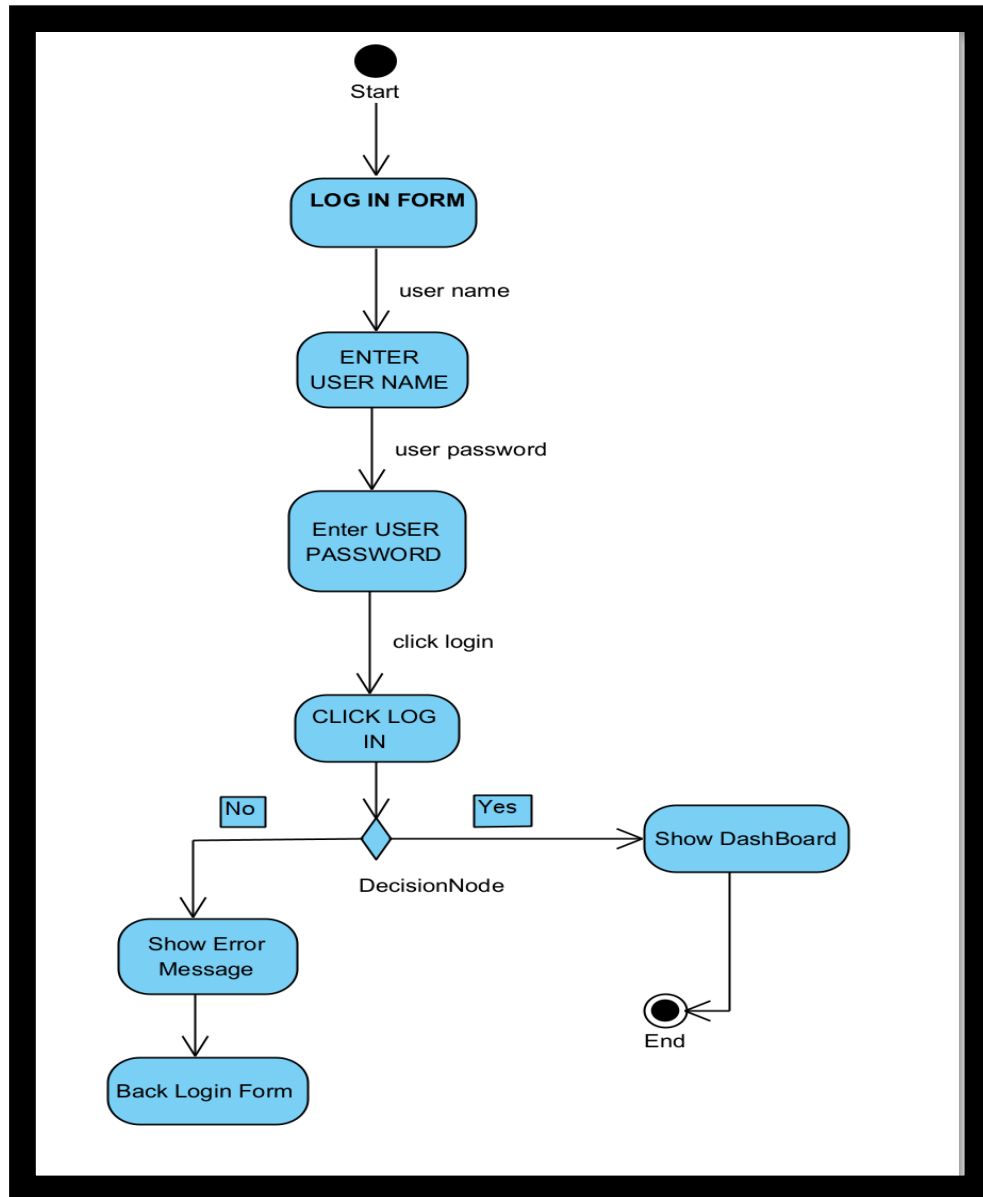
1. **Start:** The process begins with the initial node (black circle).

2. **LOG IN FORM:** The system displays the login interface to the user.
3. **ENTER USER NAME:** The user inputs their username.
4. **Enter USER PASSWORD:** The user then enters their password.
5. **CLICK LOG IN:** The user clicks the login button to submit the credentials.
6. **Decision Node (Credentials Check):**
 - ❖ The system checks if the entered username and password are valid.
 - ❖ If **Yes** (valid credentials): the user is taken to the **Show Dashboard** step, and the process ends.
 - ❖ If **No** (invalid credentials): an **Error Message** is shown.
7. **Back Login Form:** After the error message, the user is returned to the login form to try again.
8. **End:** The process concludes when the user either successfully logs in or chooses to exit.

Diagram Components:

- ❖ **Rounded rectangles** represent activities (e.g., entering username/password, clicking login).
- ❖ **Diamond** is the decision node that handles the conditional flow based on the login validation.
- ❖ **Arrows** represent the flow of control.
- ❖ **Initial node** (black circle) marks the start, and the **final node** (circle with dot) marks the end.

This diagram clearly illustrates the **decision-based flow** in the login process and helps in understanding user interactions with the system during authentication.



2 ACITIVITY DIAGRAM OF UPLOAD ASSIGNMNET

The activity diagram shown above represents the assignment upload process for a student in an academic system. It outlines the sequence of steps taken by the user to upload their assignment, along with the decision logic used to validate the uploaded file.

Explanation of Steps:

- 1. Start:** The process begins with the user opening the assignment upload functionality.
- 2. UPLOAD ASSIGNMENT:** The system opens the upload page for the user.
- 3. Select Assignment File:** The user chooses a file from their device to upload.
- 4. Upload Button:** The user clicks the upload button to submit the selected file.
- 5. File Validity:** The system checks the file for specific criteria such as format, size, or file corruption.
- 6. Decision Node – Is File Valid?:**

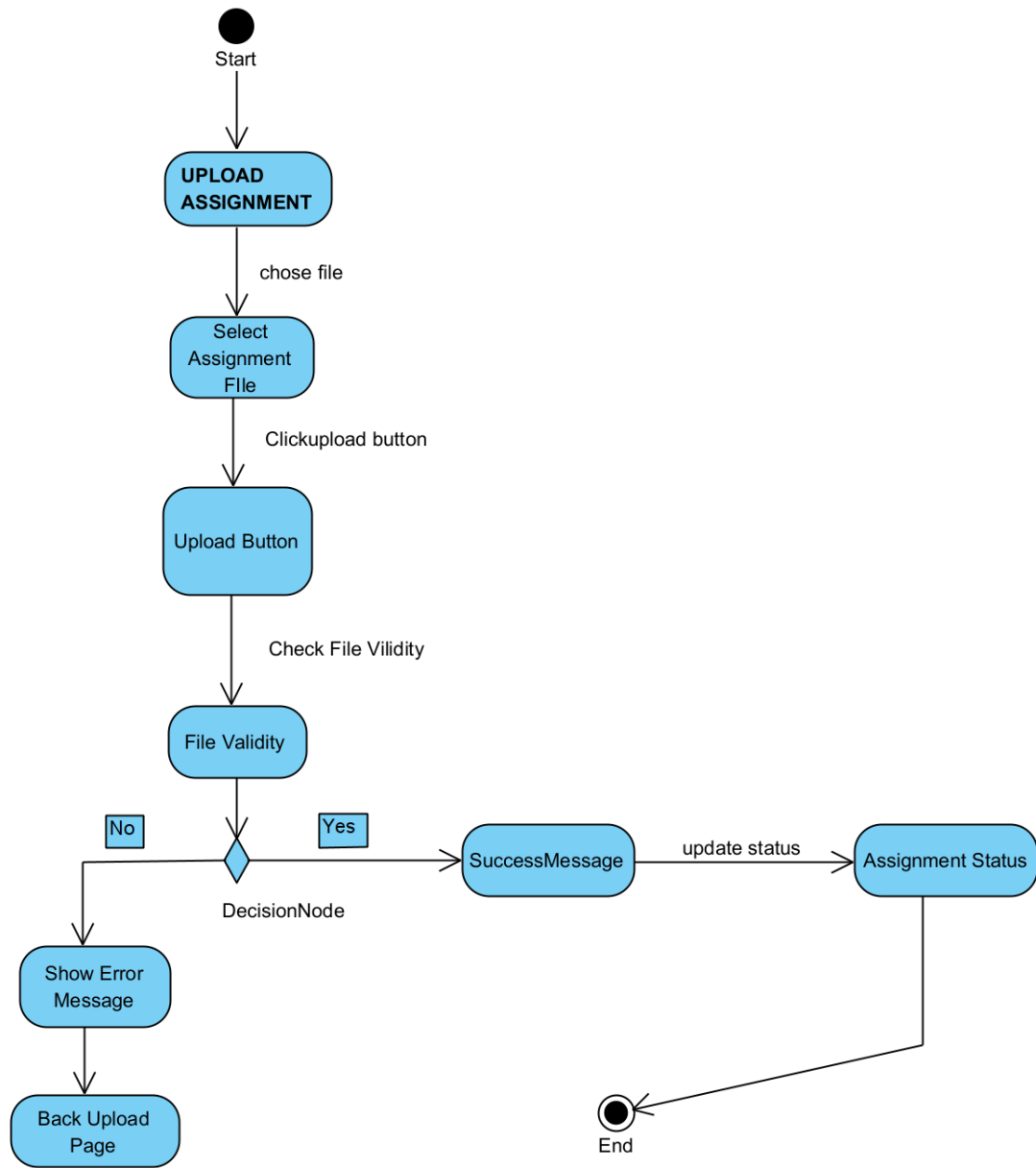
❖ **If Yes:**

- The system shows a **Success Message**.
- It then updates the **Assignment Status** (e.g., marked as "Submitted").
- The process reaches the **End**.

❖ **If No:**

- The system shows an **Error Message** indicating the issue (e.g., unsupported format or no file selected).
- The user is redirected **Back to the Upload Page** to try again.

This diagram ensures the assignment upload process is clear, user-friendly, and includes proper error handling, improving overall user experience in the system.



3 ACITIVITY DIAGRAM OF DOWNLOADING LECTURES

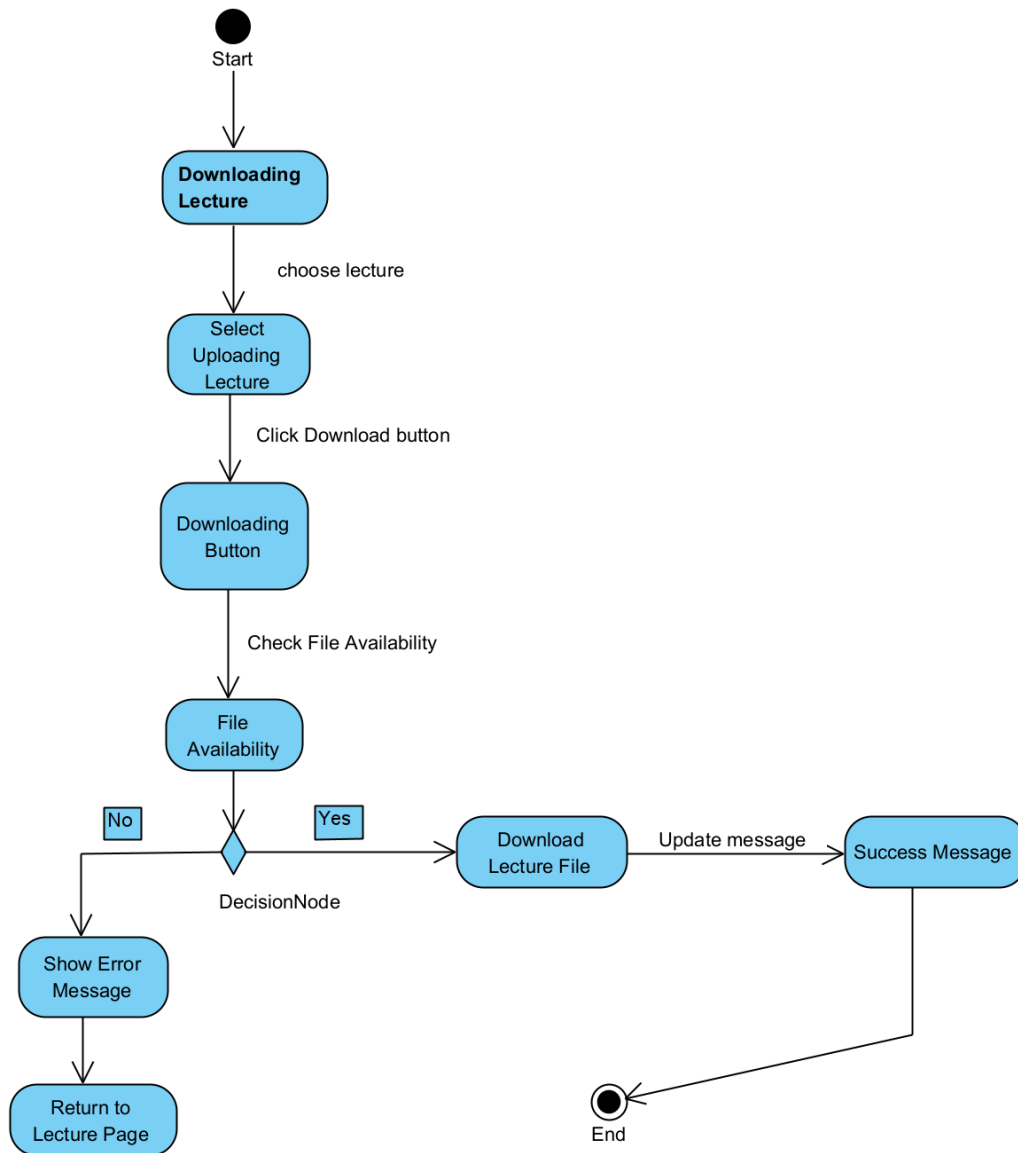
The activity diagram shown above represents the lecture download process for a student using an academic learning platform. It outlines the step-by-step flow from selecting a lecture to successfully downloading the file, with proper checks and error handling included.

Explanation of Steps:

- 1. Start:** The process begins when a student initiates the lecture download feature.
- 2. Downloading Lecture:** The system navigates to the section or page where lecture files are available for download.
- 3. Select Uploading Lecture:** The user selects the specific lecture they wish to download.
- 4. Downloading Button:** The user clicks the download button associated with the selected lecture.
- 5. File Availability:** The system checks whether the lecture file is available on the server.
- 6. Decision Node – Is File Available?:**
 - ❖ **If Yes:**
 - The system proceeds to Download Lecture File.
 - Once the file is downloaded, the system displays a Success Message confirming that the file has been saved.
 - The process then Ends.
 - ❖ **If No:**

- **The system shows an Error Message, notifying the user that the file is currently unavailable.**
- **The user is then directed to Return to Lecture Page to either retry or choose a different lecture.**
- **The process Ends after this step as well.**

This activity diagram ensures a smooth and user-friendly lecture downloading experience, guiding users through success and failure paths with clear outcomes and appropriate redirections.



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SEQUENCE DIAGRAM

1. SEQUENCE DIAGRAM FOR LOGIN PROCESS

Description:

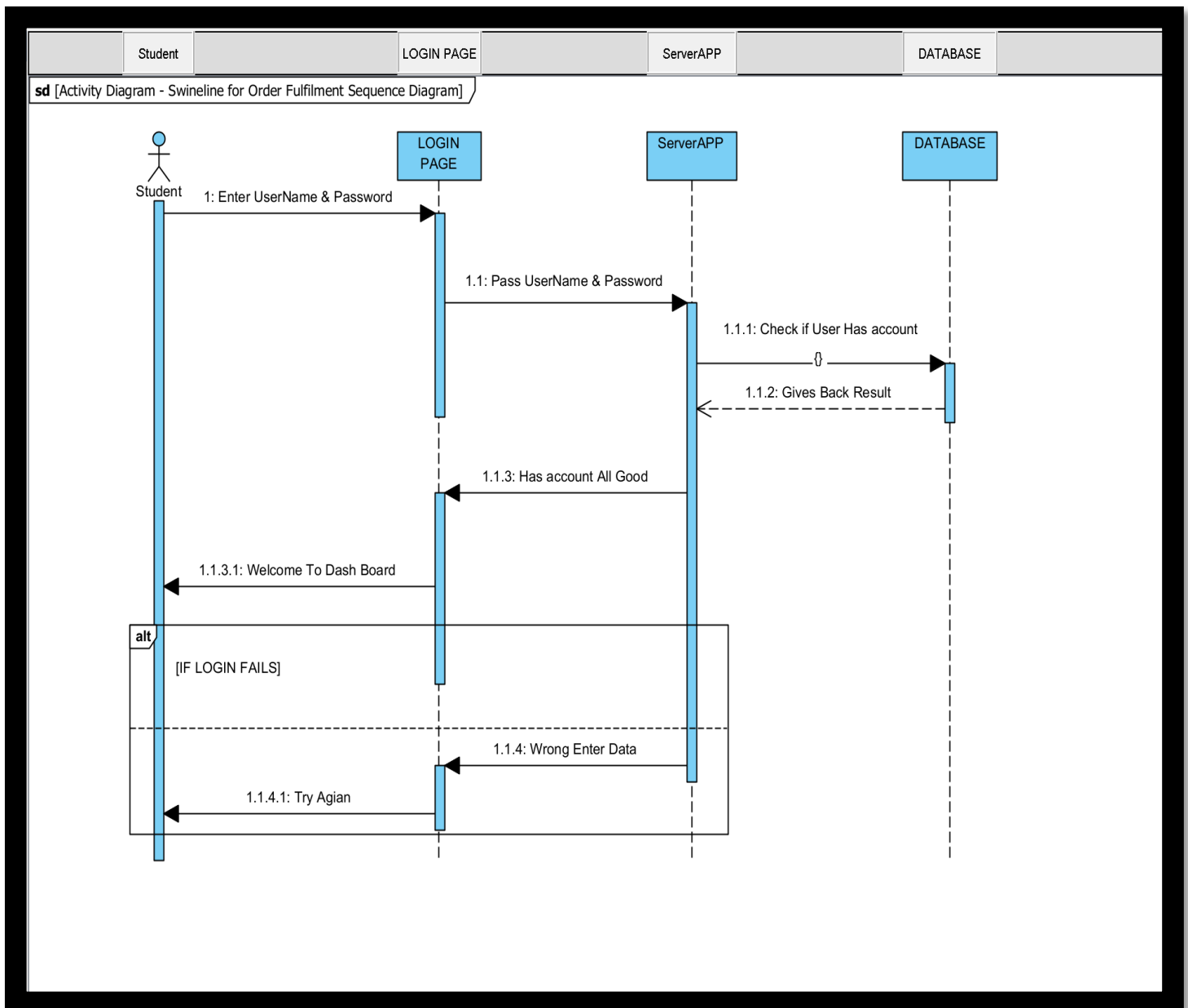
The sequence diagram shown above represents the login process for a student using an academic platform. It illustrates how the interaction flows between the user and system components during the login operation, ensuring user authentication with proper checks and feedback.

Explanation of Steps:

1. Actor (Student): The process starts when the student enters their username and password on the login page.
2. Login Page: Receives the credentials and forwards them to the Server App for verification.
3. Server App: Processes the credentials and initiates a query to the database to verify if the user exists and if the credentials match.
4. Database: Looks up the user data and returns the result (valid or invalid) to the Server App.
5. Alt Fragment – Conditional Login Outcome:
 - ❖ If credentials are valid:
 - The Server App sends a success response to the Login Page.
 - The Login Page then displays a "Welcome to Dashboard" message to the user.
 - ❖ If credentials are invalid:
 - The Server App sends an error response.

- The Login Page shows an error message like "Wrong Enter Data", prompting the user to retry.

6. End of Process: Depending on the outcome (success or failure), the user is either logged in or asked to try again.



2.SEQUENCE DIAGRAM FOR UPLOADING ASSIGNMENT

DESCRIPTION:

This sequence diagram represents the process a student follows to upload an assignment through the system. It involves interactions between the student, the upload interface (screen), the server application, the database, and the file storage system.

1. Student Action

- The student selects the assignment file and clicks the "Upload" button on the assignment upload screen.

2. Upload Request Sent

- The upload screen sends both the file metadata (such as student ID, assignment ID, file name) and the actual file content to the server application.

3. Validation

- The server application contacts the database to verify the student's identity, and checks whether the assignment submission is valid (e.g., submission is within deadline, assignment ID is valid).

4. Database Response

- The database returns the validation result (e.g., whether the student and assignment are recognized and if submission is allowed).

5. File Storage (If Valid)

- If the submission is valid, the server app stores the uploaded file in the designated file storage system (which may be a filesystem or cloud storage).

6. Storage Confirmation

- The storage system responds to confirm the file has been saved successfully.

7. Submission Recorded

- The server then records the assignment submission in the database, including details such as student ID, timestamp, and file location.

8. Database Acknowledges

- The database confirms the record has been successfully updated.

9. Success Message

- The server sends a success message back to the upload screen.

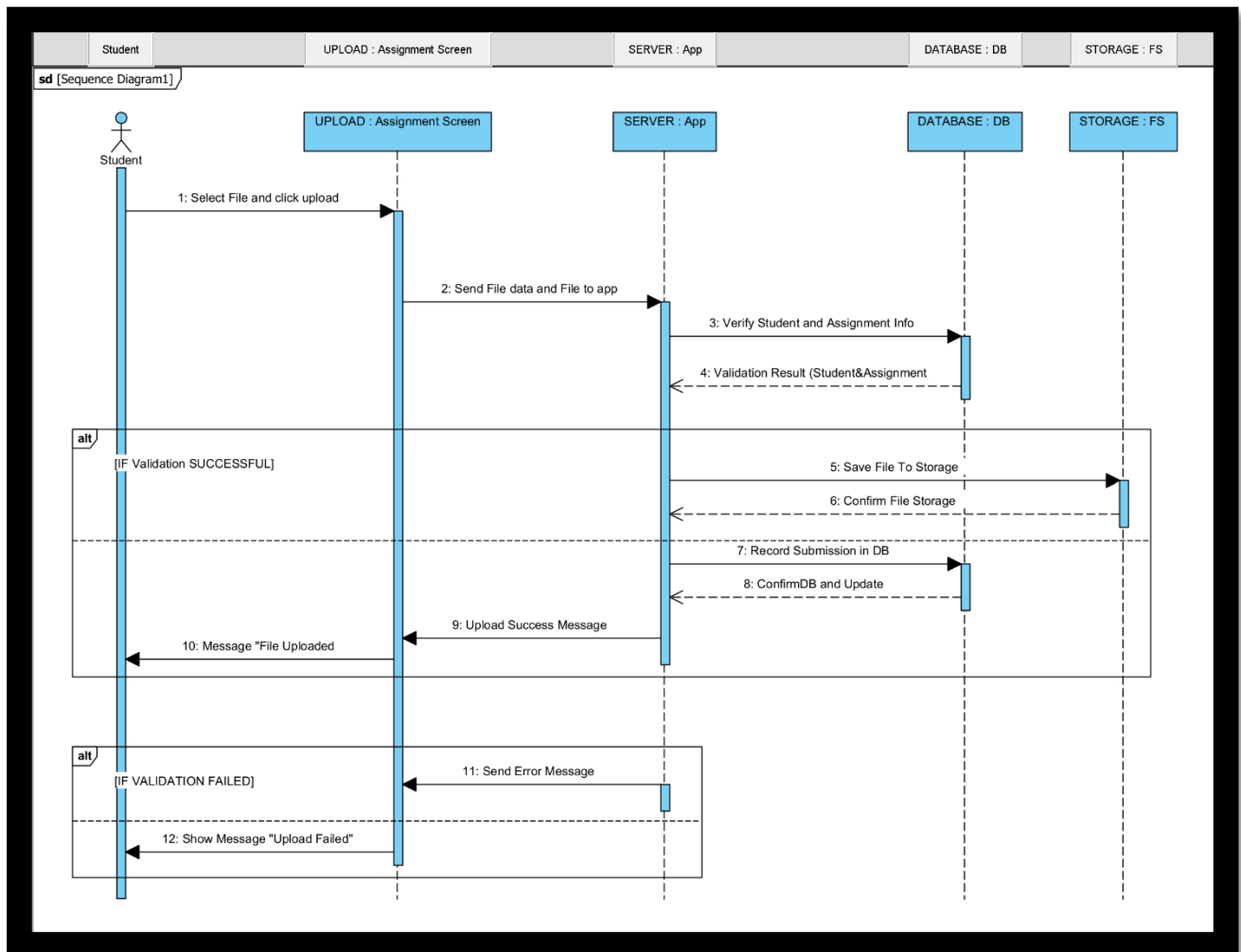
10. Final Confirmation

- The screen displays a confirmation to the student: "Your assignment was uploaded."

Error Scenario (If Validation or Upload Fails)

- If validation fails (e.g., invalid student or assignment), or if the file upload encounters an error:

- The server sends an error message back to the upload screen.
- The screen displays a message to the student: "Upload failed. Check details."



3. SEQUENCE DIAGRAM FOR DOWNLOADING LECTURE

DESCRIPTION:

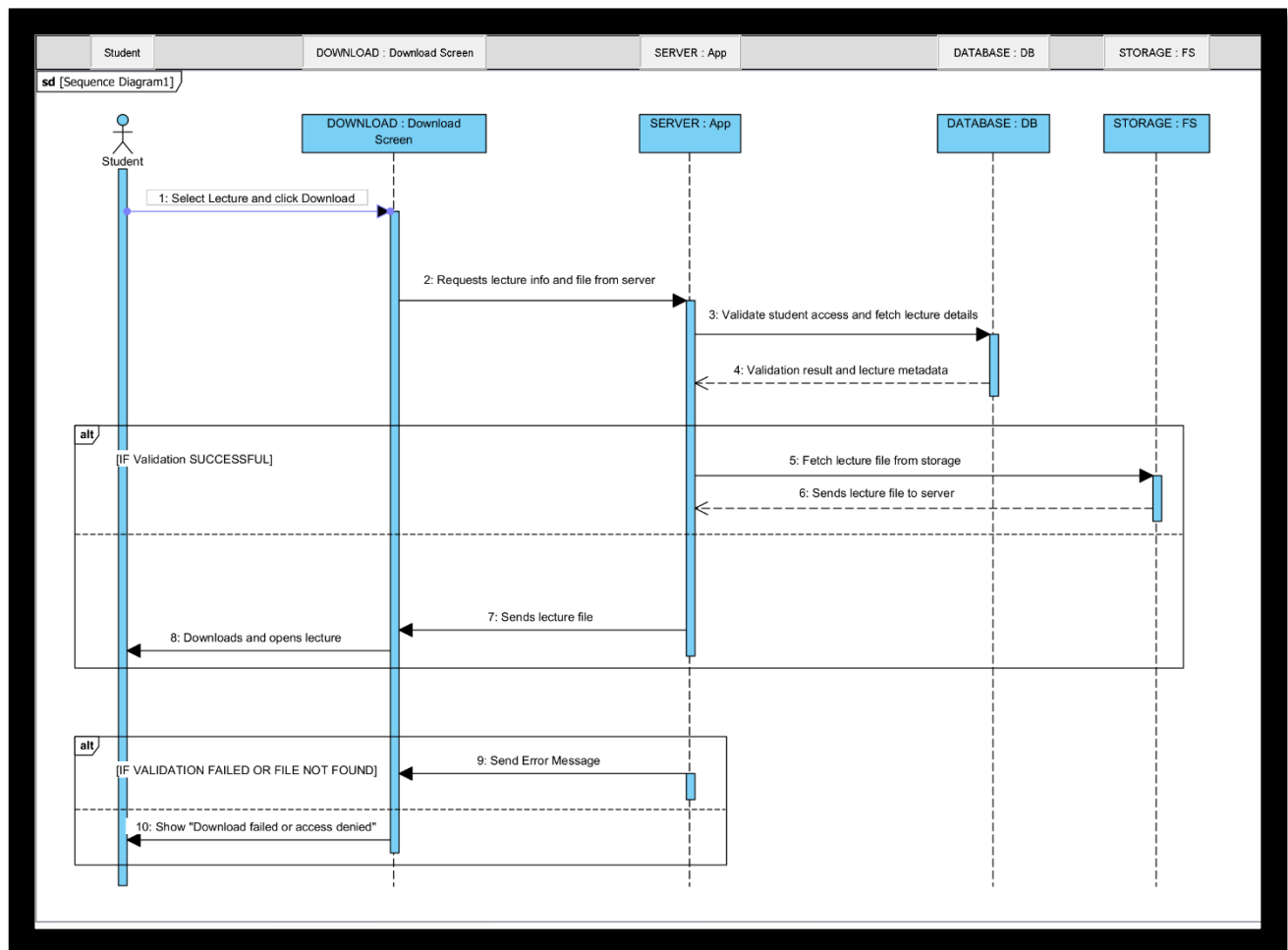
This sequence diagram illustrates the process followed by a student to download a lecture through the system. The interaction includes the student, the download screen interface, the server application, the database, and the storage system where lectures are saved.

1. Student selects a lecture from the available list on the download screen and clicks the “Download” button.
2. The Download Screen sends a request to the Server App. This request includes the lecture ID and the student’s user details.
3. The Server App contacts the Database to verify if the student is allowed to access the selected lecture (e.g., is enrolled in the course) and fetches lecture metadata like file name and location.
4. The Database responds with the validation result and metadata.
5. If access is valid:
 - The Server App sends a request to the Storage system to retrieve the actual lecture file.
 - The Storage system locates and sends the file back to the Server App.
6. The Server App forwards the file to the Download Screen.
7. The Download Screen starts the download, and the student receives and opens the lecture file.

Alternate Flow (If Access Fails or File Missing)

- If the student is not authorized or the file is missing:

- ❖ The Server App sends an error message back to the Download Screen.
- ❖ The Download Screen shows an error, like: “Download **failed or access denied.**”



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